Solar-sail ocean drone, 1.5 m

Drawing set for the custom-built parts, with 3D-print files and full-size templates

Sheet	Title
D00	Cover, index and general notes
D01	General arrangement
D02	Hull lines and table of offsets
D03	Interior arrangement and systems
D04	Keel fin and lead bulb
D05	Rudder, stock and steering
D06	Wing sail and tail
D07	Mast step, spar and stern stub
D08	Deck hatch, solar layout, penetrations
D09	Printed parts P01, P02
D10	Printed parts P03, P04
D11	Printed parts P05, P06
D12	Printed parts P07, P08
D13	Printed parts P09, P10
D14	Printed parts P11, P12
D15	Printed parts P13, P14
D16	Printed parts P15, P16
D17	Printed parts P17

Part	3D-printed part	Qty	Material / settings
P01	Wind-vane sensor housing	1	ASA or PETG, 100% walls 4
P02	Wind-vane arm with fin and counterweight pocket	1	ASA or PETG
P03	Sensor cap and stub-tube socket	1	ASA or PETG
P04	Magnet holder for vane shaft	1	PETG
P05	Antenna platform clamp	1	ASA
P06	Transom bracket for 20 mm stub tube	1	ASA, 6 walls
P07	Mast bearing housing	1	PETG or ASA, 100% infill
P08	Spar clamp with tail-boom socket	1	ASA or PETG, 100% infill
P09	Tail servo housing (boom end)	1	ASA or PETG
P10	Tail fin, lower and upper halves	1	PETG or ASA, 0.2 mm layers
P11	Rudder tiller arm	1	PETG, 100% infill
P12	Rudder servo mount	1	PETG or ASA
P13	Rudder blade core, upper and lower halves	1	PETG, 15% gyroid
P14	Keel bulb casting pattern, forward and aft halves	1	PLA or PETG, 3 walls
P15	Electronics tray	1	PETG
P16	Battery strap anchor	4	PETG
P17	Penetrator boss pad	4	ASA or PETG, 100% infill

General notes

1. All dimensions in millimetres. Model origin: x aft from the bow, z up from the hull bottom at midships, y to port.
2. Design waterline (DWL) z = 90 at about 21 kg all-up mass (hull 19.4 kg displacement + appendages).
3. Printed parts: 0.4 mm nozzle, 0.2 mm layers, 4+ walls. Use ASA outdoors in UV; PETG is fine below deck.
   STL files are in print orientation; no supports needed except where a sheet says so.
4. Bought-in parts follow the Recommended Parts list in the build plan. Servo tab spacing, AS5600 board size and
   patch-antenna footprint vary by supplier: printed parts use slots and recesses rather than fixed hole patterns.
5. Full-size cutting templates (hull frames, foils, wing, hatch) are in Templates.pdf (tiled on Letter, 1:1)
   and as DXF files. Check the 100 mm scale bar on every printed page before cutting.
6. All fasteners and shafts 316 stainless. Bed every hull penetration in polyurethane sealant.
7. This is a hobby design, not a classed or certified vessel. Tank-test self-righting before going to sea.

SOLAR-SAIL OCEAN DRONE 1.5 m

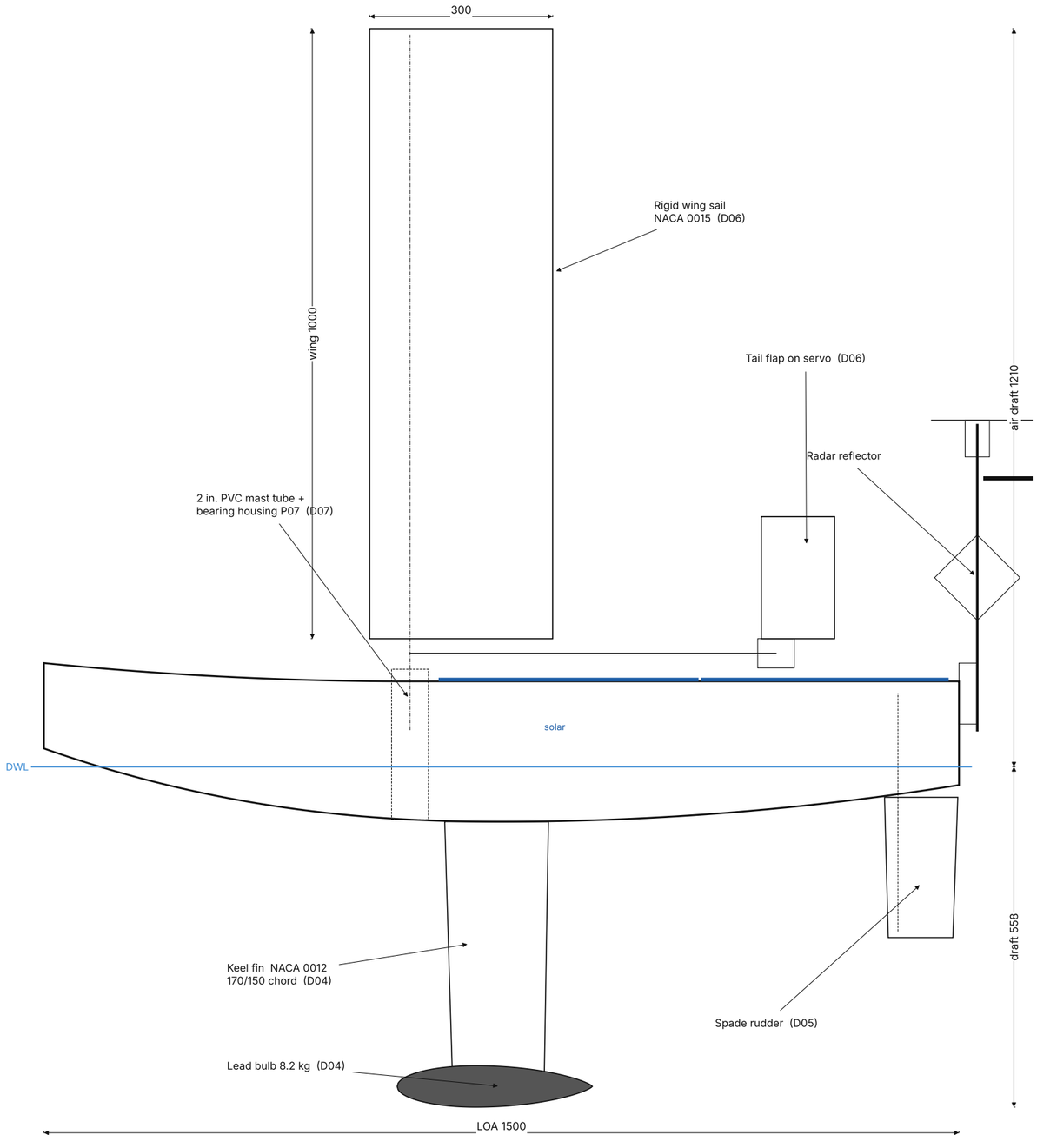
COVER, INDEX AND GENERAL NOTES

Units: mm

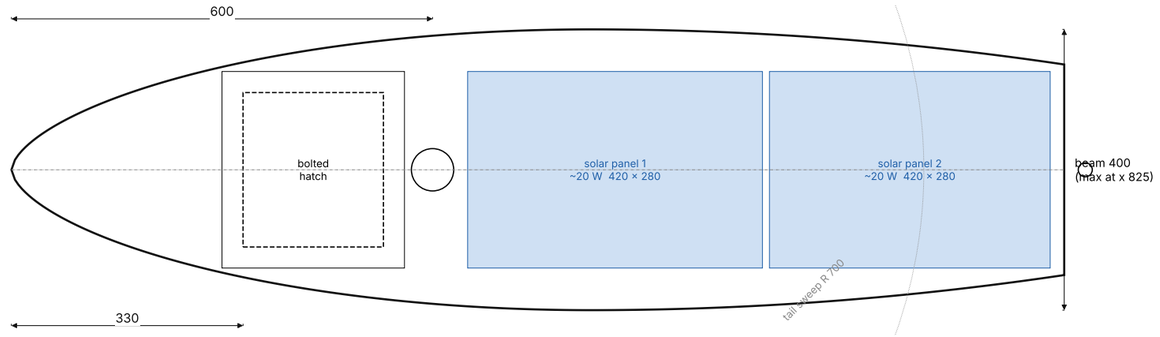
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET **D00**

REV **A**



PROFILE (starboard side, wing feathered aft)



PLAN (deck)

Key data

Length overall 1500 Beam 400 Hull depth 230
Draft 558 (DWL to bulb bottom) Air draft 1210
Displacement about 21 kg Ballast 8.2 kg lead
Wing 1000 × 300 (0.30 m2), free-rotating on 25 mm spar
Solar about 40 W on deck Battery 192 Wh LiFePO4
Self-righting: verify in a tank test (Gate 3)

SOLAR-SAIL OCEAN DRONE 1.5 m

GENERAL ARRANGEMENT

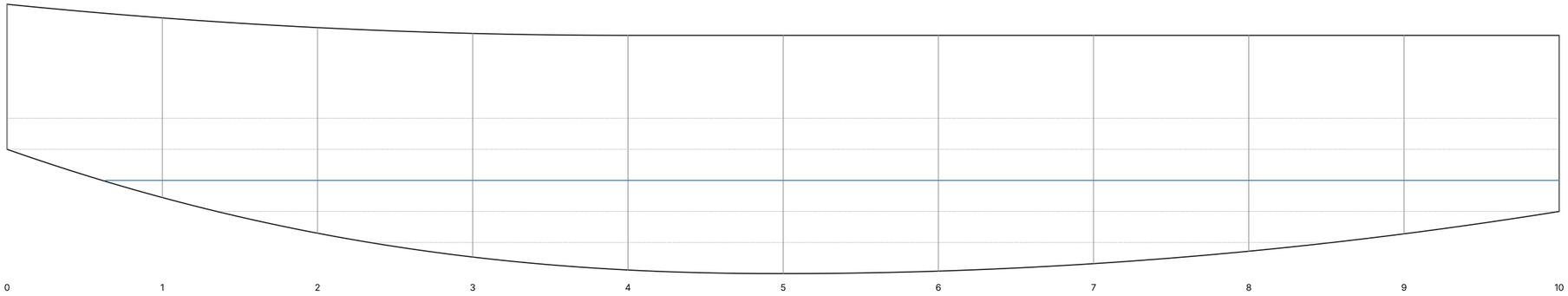
Units: mm Scale 1:8.5 on tabloid

Date 2026-11-01 Drawn with Claude Check all dims before cutting

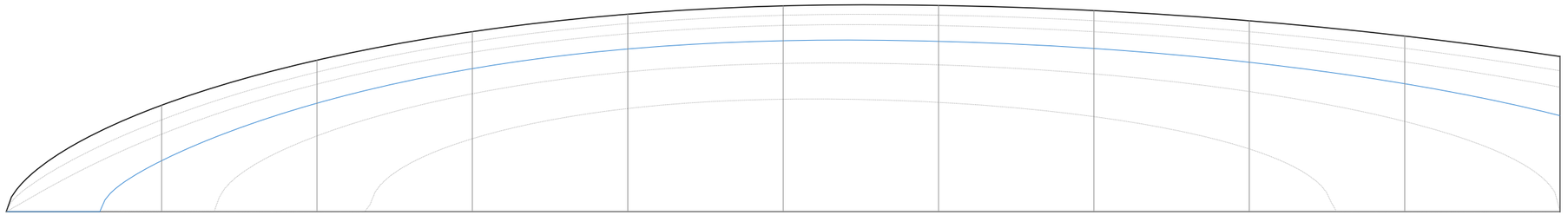
SHEET **D01**

REV **A**

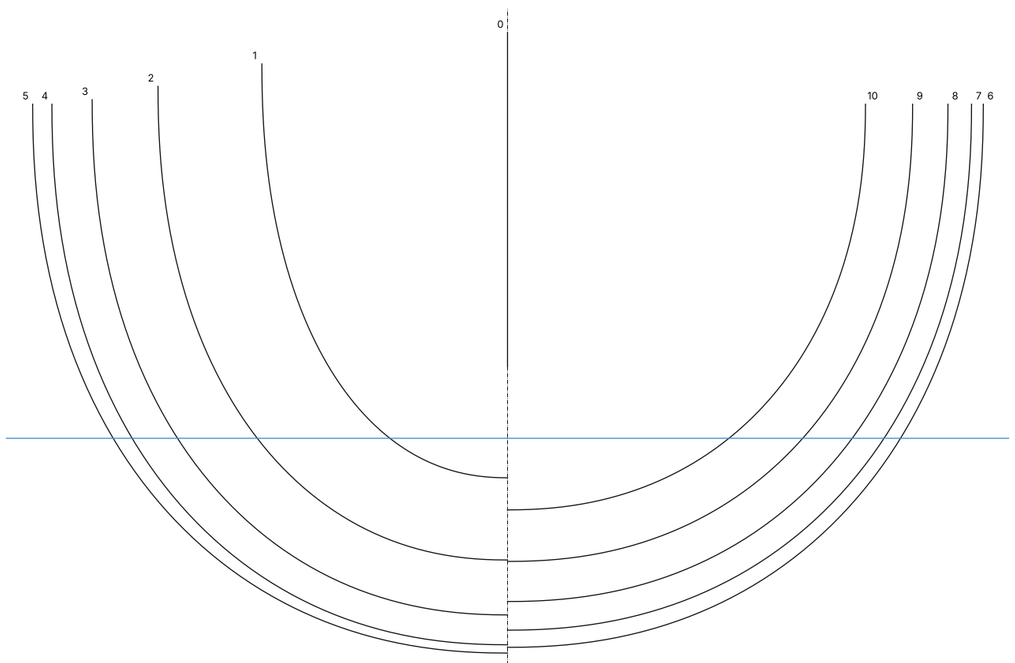
PROFILE



HALF-BREADTH PLAN (deck edge solid, DWL blue, waterlines z 30/60/120/150 dotted)



BODY PLAN (stations 0-5 left, 5-10 right)



Stn	x	keel z	deck z	1/2 b deck	1/2 b z30	1/2 b z60	1/2 b DWL	1/2 b z150
0	0	120.0	260.0	0.0	0.0	0.0	0.0	0.0
1	150	73.4	246.9	102.9	0.0	0.0	49.3	88.8
2	300	39.0	237.5	146.5	0.0	73.3	104.9	134.9
3	450	16.0	231.9	174.1	70.6	114.0	138.3	164.4
4	600	3.5	230.0	190.9	99.7	135.2	157.3	181.9
5	750	0.0	230.0	199.0	108.8	143.3	165.3	189.9
6	900	2.4	230.0	199.4	105.7	142.0	164.7	190.0
7	1050	9.6	230.0	194.4	91.9	133.2	157.8	184.6
8	1200	21.6	230.0	184.6	60.7	116.2	144.4	174.0
9	1350	38.4	230.0	169.8	0.0	87.3	123.7	158.0
10	1500	60.0	230.0	150.0	0.0	0.0	92.8	136.3

Section formula (exact; use it for intermediate points)

Half-breadth at height z: y = b(x) [1 - (1 - t)^2.2]^(1/2.2), t = (z - keel z) / (deck z - keel z)
b(x) = 200 [1 - (1 - x/825)^2]^0.6 forward of x 825; 200 - 50 ((x - 825)/675)^2 aft
Keel z = 120 (1 - x/750)^2.2 forward of x 750; 60 ((x - 750)/750)^2 aft
Deck z = 230 + 30 (1 - x/600)^2 forward of x 600; 230 aft

Construction: 10 mm H80 foam strips over 11 MDF frames (Templates T01-T11, cut on the dashed frame line = hull minus 11 mm), 2 × 6 oz glass outside, 1 × 6 oz inside, peel ply.

SOLAR-SAIL OCEAN DRONE 1.5 m

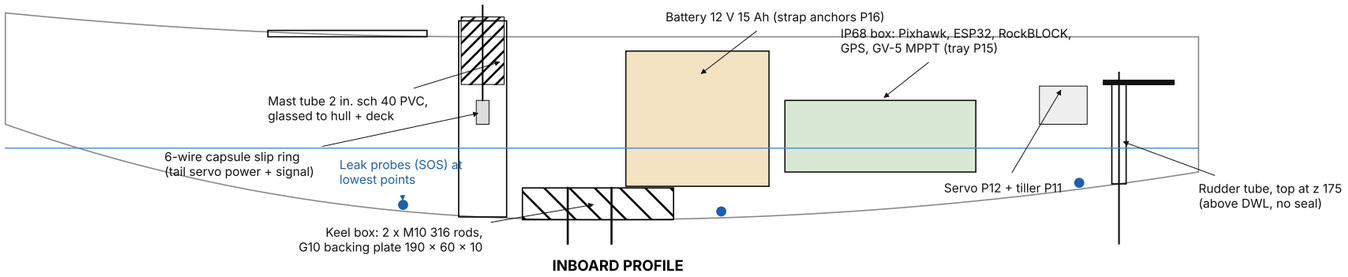
HULL LINES AND TABLE OF OFFSETS

Units: mm Profile & plan 1:5, body plan 1:2.5

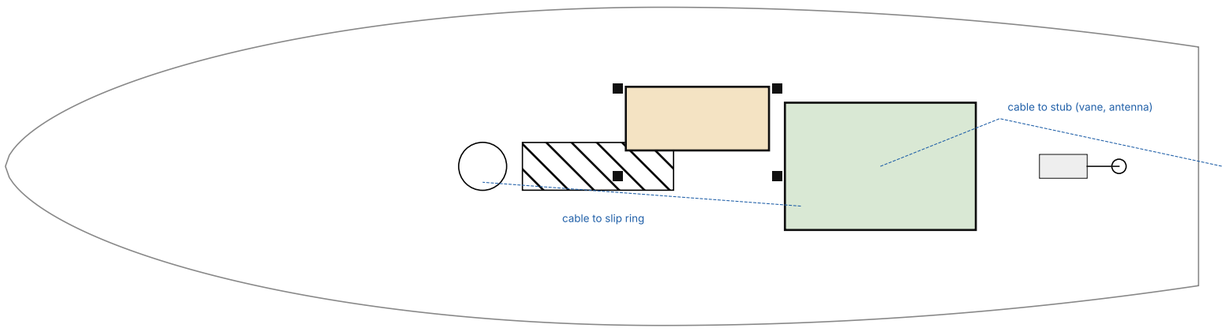
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET **D02**

REV **A**



INBOARD PROFILE



INTERIOR PLAN (below deck)

Systems notes

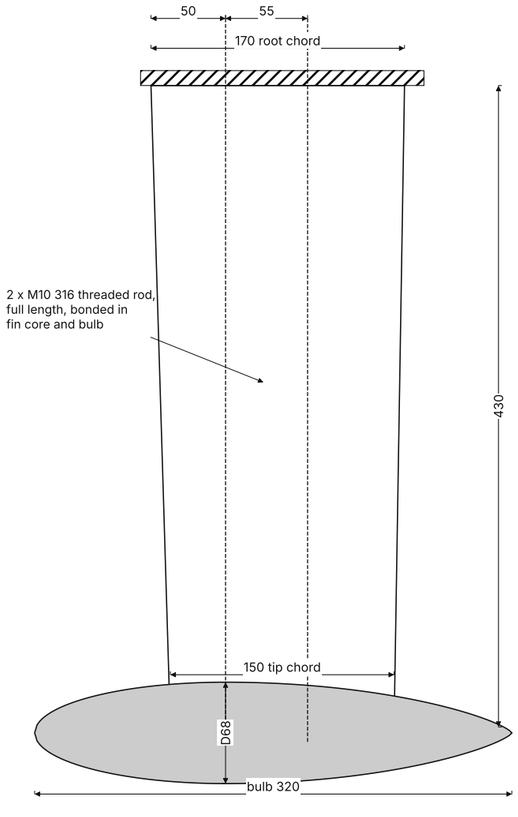
Masses (approx, kg): hull + deck 4.5, fin 1.2, bulb 8.2, rudder 0.3, wing + spar + tail 1.6, battery 2.0, electronics + cables 1.2, solar 1.0, misc 1.0 → about 21 kg.
Battery sits beside the keel box to keep weight low and on the centreline; the IP68 box is aft of it so cables to the stub and rudder stay short.
All penetrations use WetLink M10 on P17 pads. Solar cables enter through one penetrator per panel just inside the panel edge.

SOLAR-SAIL OCEAN DRONE 1.5 m

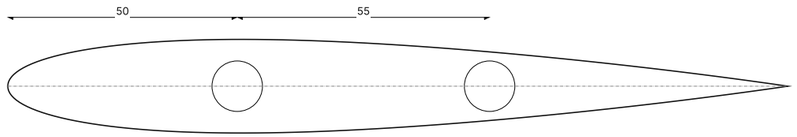
INTERIOR ARRANGEMENT AND SYSTEMS

Units: mm Scale 1:7.5 Date 2026-11-01 Drawn with Claude Check all dims before cutting	SHEET **D03**	REV **A**

from nose	diameter
0	0.0
20	36.5
40	49.4
60	57.6
80	63.0
100	66.4
120	67.9
140	67.8
160	66.6
180	64.2
200	60.7
220	55.9
240	49.8
260	42.1
280	32.5
300	20.1
320	0.0



FIN + BULB ELEVATION



FIN ROOT SECTION NACA 0012 (1:1.3)

Fin construction

1. Hot-wire cut a blue XPS or Divinycell core with templates T12 (root) and T13 (tip).
2. Bore 2 × 12 mm holes at 50 and 105 from the root LE; bond the M10 rods in with thickened epoxy.
3. Glass 3 × 6 oz each side, +/-45 deg on the inner layer; fair, then 2-part epoxy primer.
4. Rods pass 40 mm into the keel box and take 316 nuts on a G10 plate (190 × 60 × 10) glassed into the hull.

Bulb
Volume 721 cm3 → 8.2 kg of lead. Make a plaster mould from pattern P14a/b
(printed), cast outdoors with PPE, then drill 2 × 12 mm holes 60 deep at the rod positions
(91 + 37 and 91 + 92 mm from the nose, measured on the slot) and bond the rods in with epoxy.
Fair with epoxy filler; seal with 2 coats epoxy barrier coat.

SOLAR-SAIL OCEAN DRONE 1.5 m

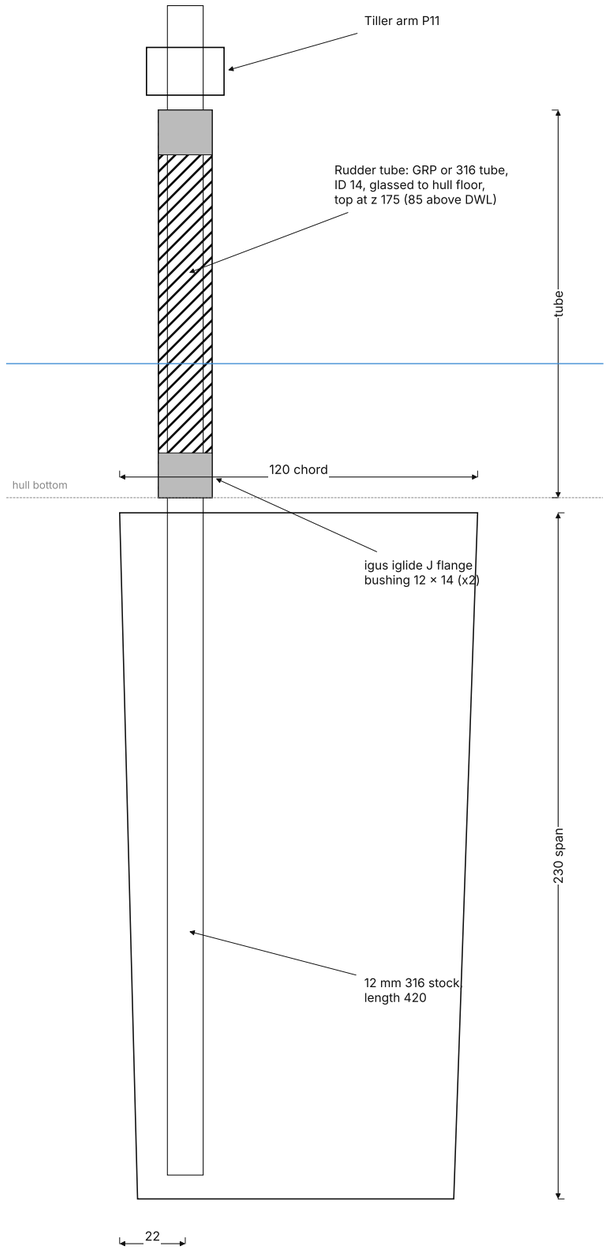
KEEL FIN AND LEAD BULB

Units: mm Scale 1:4

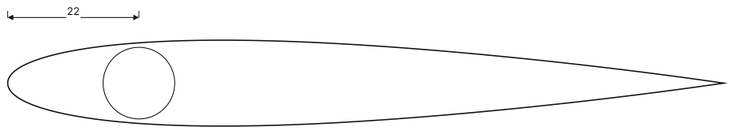
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D04

REV A



RUDDER ELEVATION (looking to port)



BLADE SECTION NACA 0012 (1:1)

Build

1. Print cores P13a/b, slide on the stock, pin with 4 mm 316 pins, epoxy the joint.
2. Glass 2 × 6 oz over the blade, wrapping the tip; fair to template T14.
3. Balanced rudder: about 15% of the area is ahead of the stock, so the servo needs little holding torque and can be powered down between corrections.
4. Stock: 12 mm 316 round bar, 420 long; 3 mm cross-hole for the tiller pin.
5. Bottom bushing sits flush with the hull bottom; top bushing at the tube top.
6. Stop the rudder at +/-35 deg with the servo end points (ArduPilot SERVO1_MIN/MAX).

SOLAR-SAIL OCEAN DRONE 1.5 m

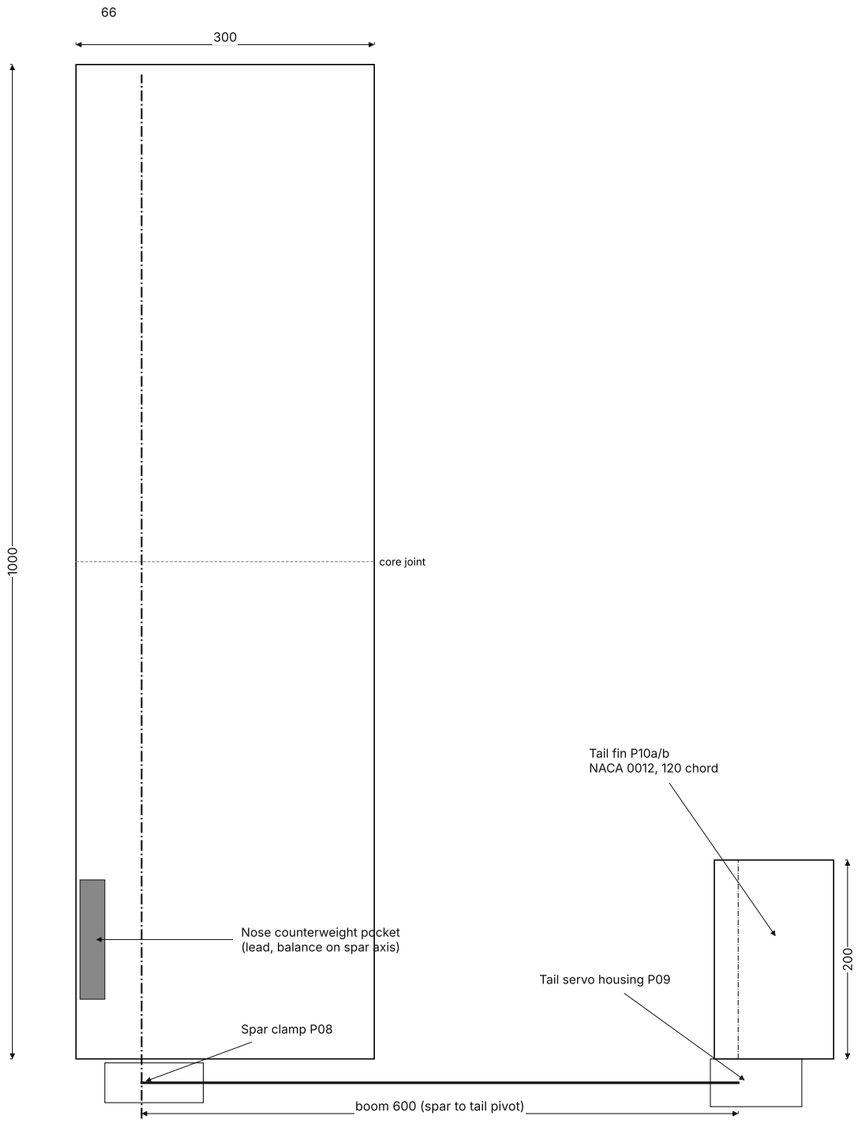
RUDDER, STOCK AND STEERING

Units: mm Scale 1:2

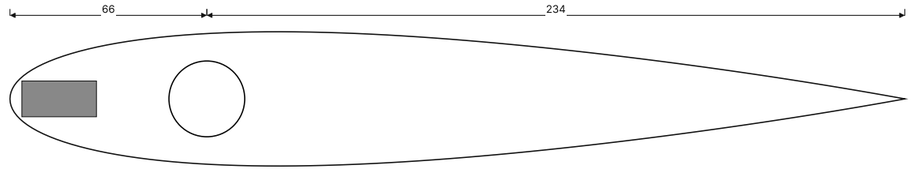
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET **D05**

REV **A**



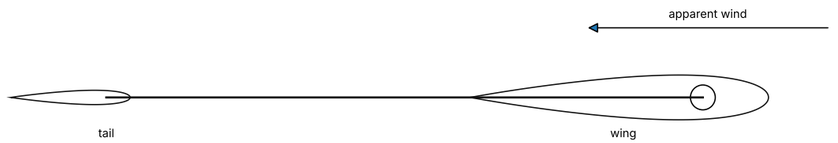
WING + TAIL ELEVATION



WING SECTION NACA 0015 (1:2)

Wing build

Core: two 500 mm XPS blocks hot-wire cut with template T15 (same root and tip), bored 25.4 mm for the spar.
Skin: 2 × 6 oz glass, vacuum bagged if possible. Root and tip: 3 mm ply caps. Spar: 25 mm OD carbon tube, 1100 long.
Balance the wing+tail assembly so its centre of mass is on the spar axis (lead in the nose pocket).
Tail angle sets the wing's angle of attack: about +/-8 deg tail gives the working range; 0 deg = wing feathered.



PLAN OF ROTATING RIG (1:6)

SOLAR-SAIL OCEAN DRONE 1.5 m

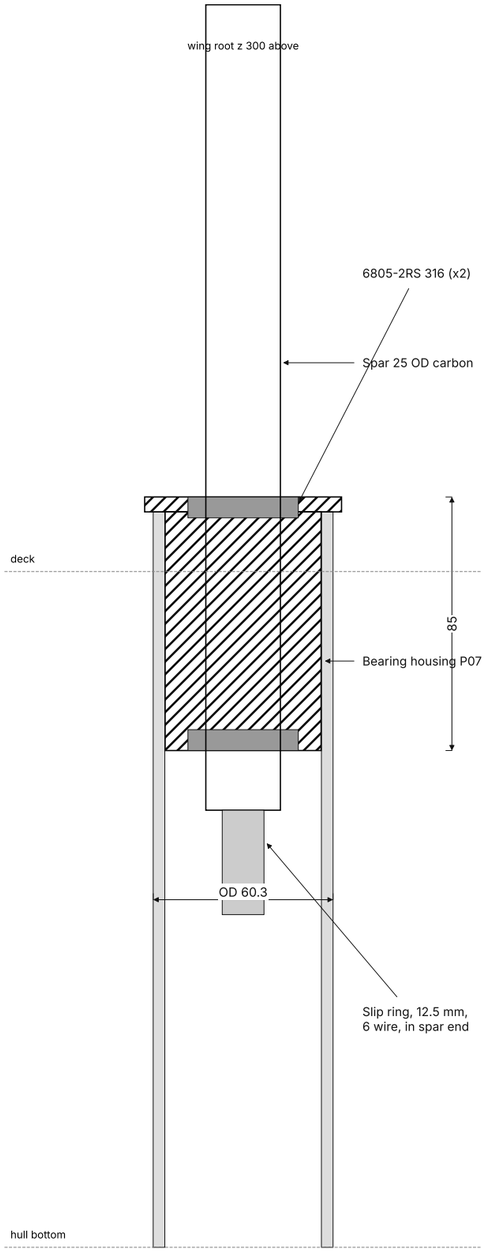
WING SAIL AND TAIL

Units: mm Elevation 1:6, plan 1:6, section 1:2

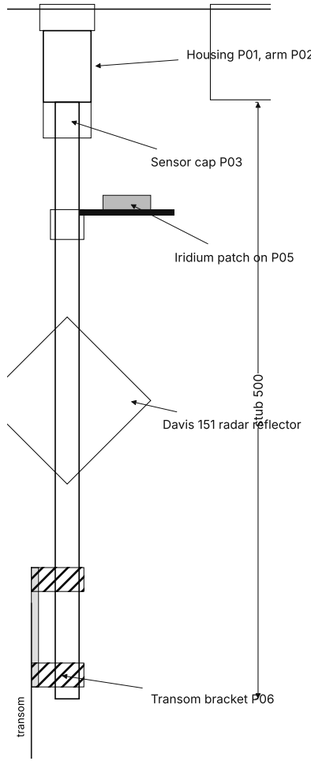
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET **D06**

REV **A**



MAST STEP SECTION (1:2)



STERN STUB (1:5)

Notes

Spar: 25 mm OD carbon tube, 1150 long (z 150 to 1300).
Slip ring in the spar's lower end carries 3 wires to the
tail servo; it must turn freely through 360 deg.
Lock the spar against lifting with a 316 collar under
the lower bearing.

Stub: 20 mm OD x 1.5 wall 6061 aluminium (anodised) or
carbon tube, 500 long; cables run inside it to the
vane cap and down through a WetLink in the transom.
Vane shaft: 8 mm 316, 75 long, 2 x S608-2RS bearings.

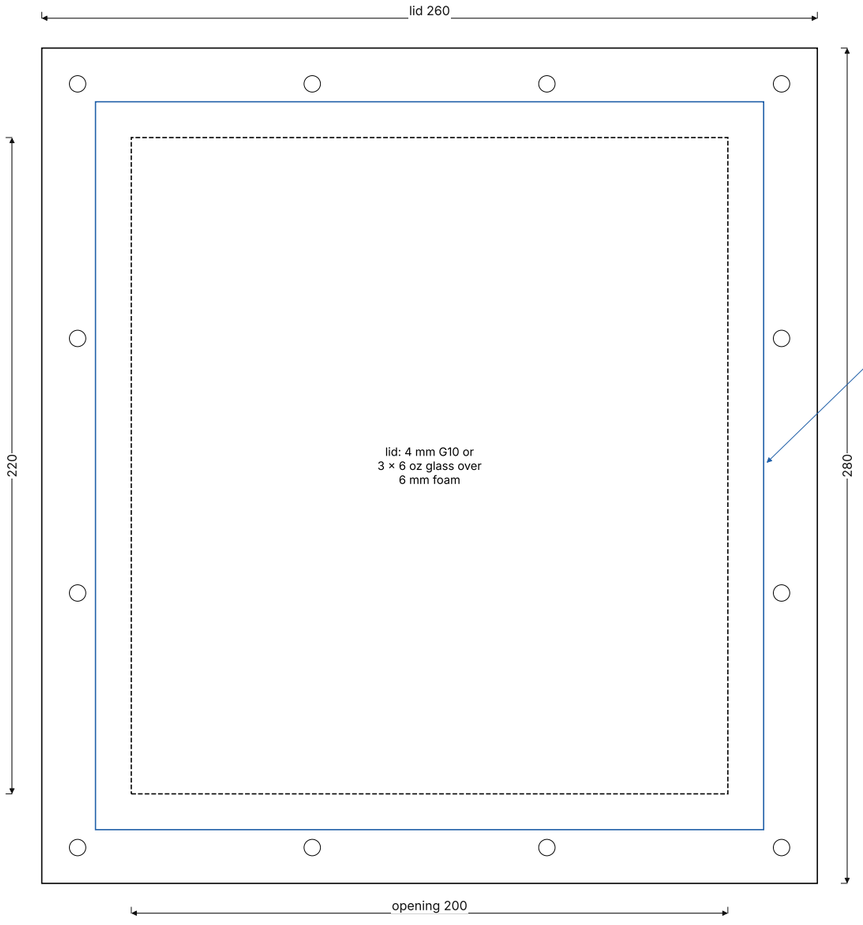
SOLAR-SAIL OCEAN DRONE 1.5 m

MAST STEP, SPAR AND STERN STUB

Units: mm Scale 1:2 and 1:5

Date 2026-11-01 Drawn with Claude Check all dims before cutting

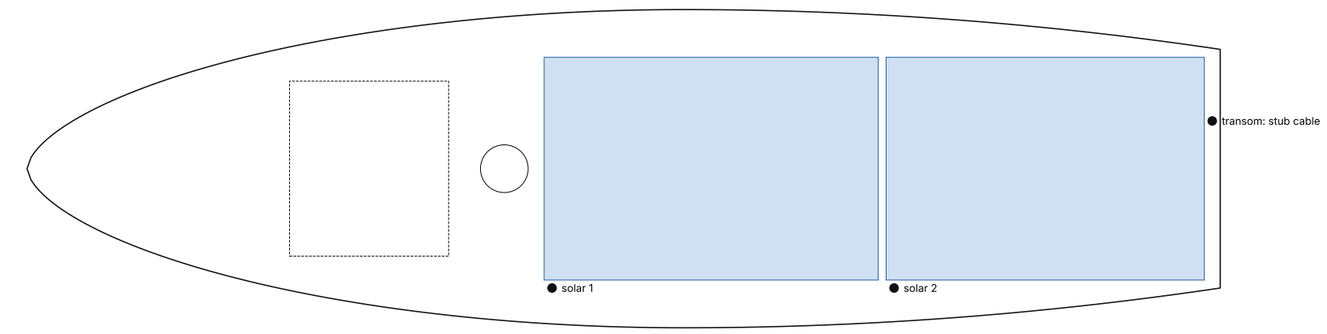
SHEET	REV
D07	A



HATCH LID AND OPENING (1:2)

Hatch

12 x M5 316 bolts into bonded-in M5 inserts (or nuts glassed on a doubler).
Doubler under the deck: 30 mm wide frame of 6 mm foam + 2 × 6 oz glass.
Torque evenly; grease the O-ring with silicone grease; tape the seam for the crossing.
Template T16 gives the opening, bolt holes and lid outline at 1:1.



DECK PENETRATIONS (P17 pads + WetLink)

SOLAR-SAIL OCEAN DRONE 1.5 m

DECK HATCH, SOLAR LAYOUT AND PENETRATIONS

Units: mm Hatch 1:2, deck 1:7.5

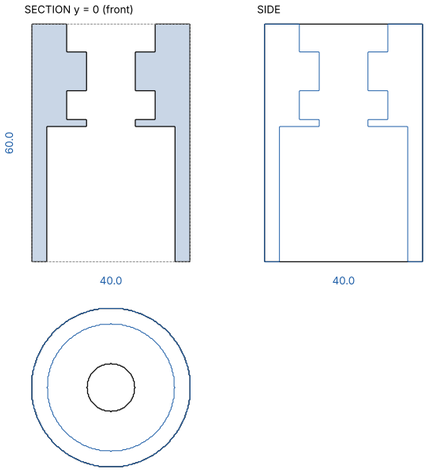
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D08

REV A

P01 Wind-vane sensor housing

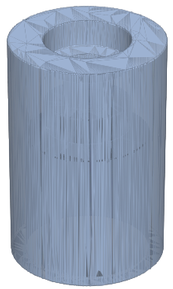
File: stl/P01_vane_housing.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
OD	40
Height	60
Bearing seats	2 x D22.1 × 7
Cavity	D32 × 34

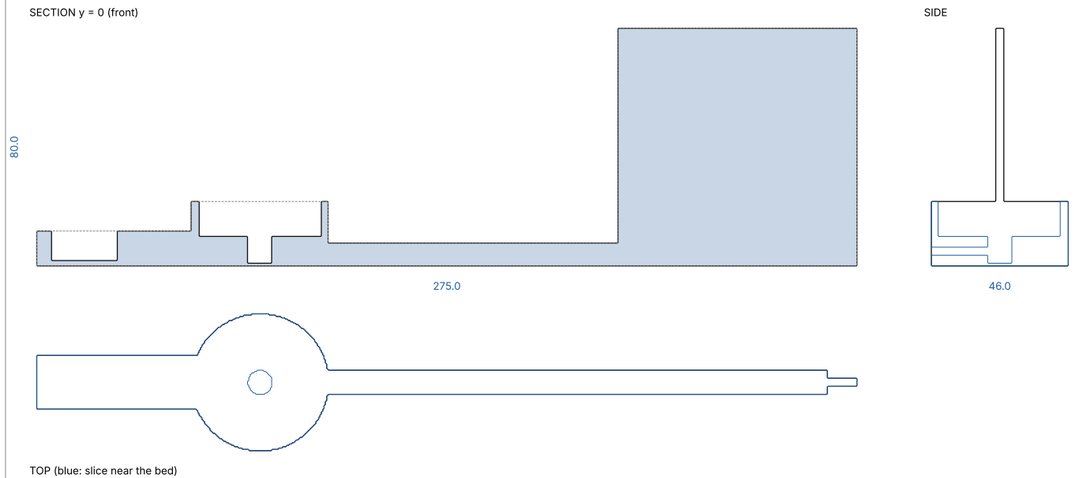
Material: ASA or PETG, 100% walls 4

- Two S608-2RS stainless bearings (8×22×7) press in from the top.
- AS5600 board sits on cap P03 inside the cavity; magnet on shaft end faces it.
- Bottom cavity is closed by cap P03, bonded with thickened epoxy.



P02 Wind-vane arm with fin and counterweight pocket

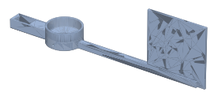
File: stl/P02_vane_arm.stl Views at 1:2 Print orientation: z up from the bed



Feature	Size
Arm length	190 + 75
Fin	80 × 80 × 3
Hub	D46 × 22
Shaft bore	D8.1 × 9 blind

Material: ASA or PETG

- Printed upside-down: z=0 face is the top of the hub in service.
- Skirt (D41 inside) overlaps the housing top as a rain labyrinth.
- Fill the pocket with lead shot + epoxy until the arm balances on the shaft.
- M3 set screw (tap the D2.5 radial hole) locks the hub on the 8 mm shaft.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

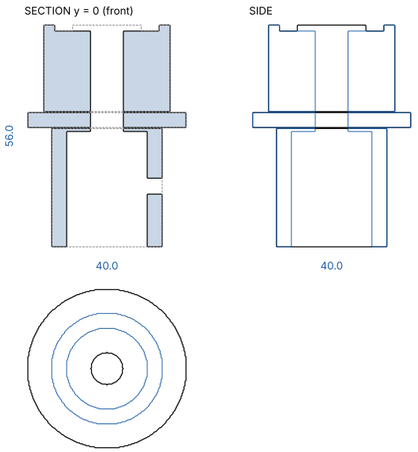
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET **D09**

REV **A**

P03 Sensor cap and stub-tube socket

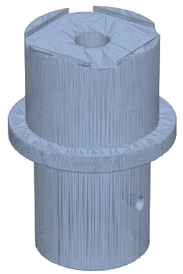
File: stl/P03_sensor_cap.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Spigot	D31.6 × 22
Flange	D40 × 4
Socket	OD28, ID20.4 × 30

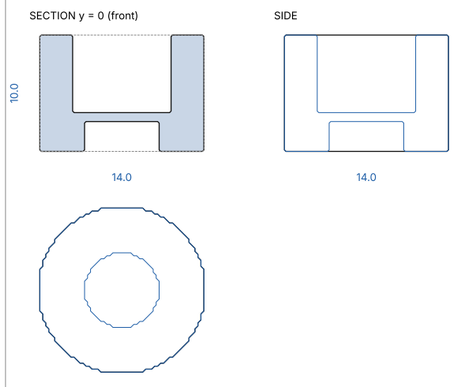
Material: ASA or PETG

- Spigot (D31.6) slides into the P01 cavity; bond and seal with epoxy.
- Recess on top takes the AS5600 board (glue down); cable runs down the D8 hole into the 20 mm stub tube.
- Socket fits the 20 mm stub tube; M4 set screw (D3.3 tap hole).
- Printed socket-down as modelled.



P04 Magnet holder for vane shaft

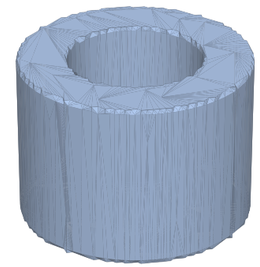
File: stl/P04_magnet_holder.stl Views at 1:0.5 Print orientation: z up from the bed



Feature	Size
OD	14
Height	10
Magnet pocket	D6.2 × 2.7
Shaft bore	D8.05 × 6.5

Material: PETG

- Press onto the bottom end of the 8 mm 316 shaft below the lower bearing.
- Glue a 6 × 2.5 mm diametric magnet (AS5600 type) into the bottom pocket.
- Also acts as the retaining collar that stops the shaft lifting out.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

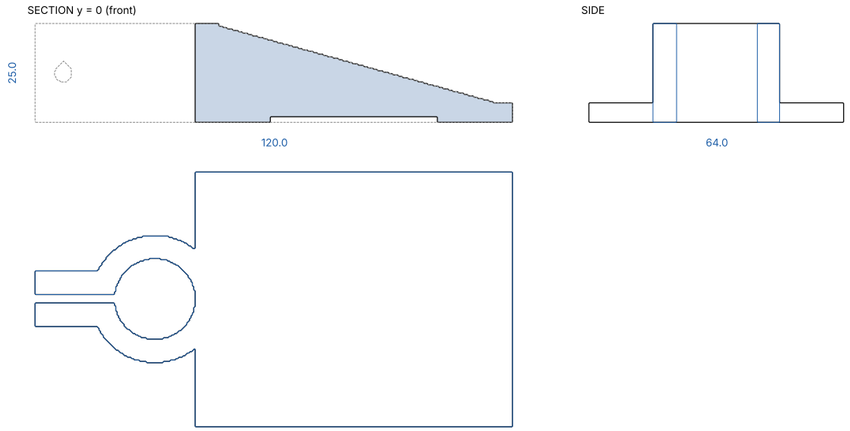
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D10

REV A

P05 Antenna platform clamp

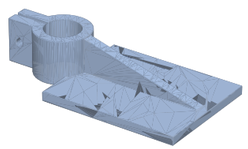
File: stl/P05_antenna_platform.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Platform	80 × 64 × 5
Collar	OD32, ID20.3 × 25

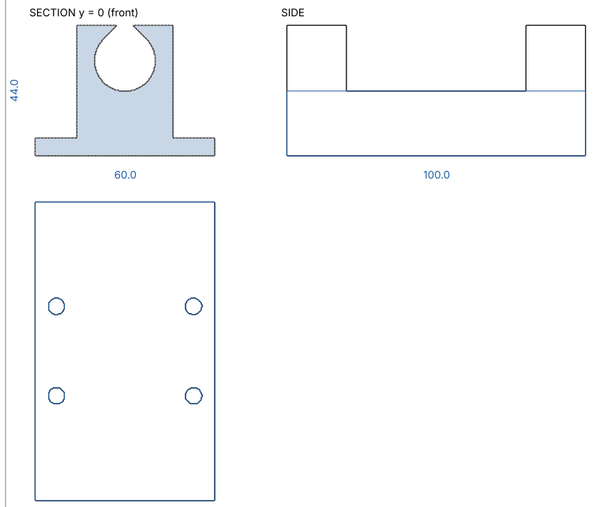
Material: ASA

- Clamps to the 20 mm stub tube at 560 mm above the hull bottom.
- Platform face printed on the bed; the D42 × 1.5 pocket takes a steel fender washer for the magnet-mount patch antenna.
- Squeeze slot closed with an M4 bolt (D4.3 hole) through the ears.



P06 Transom bracket for 20 mm stub tube

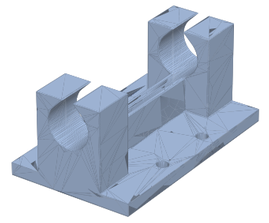
File: stl/P06_transom_bracket.stl Views at 1:2 Print orientation: z up from the bed



Feature	Size
Plate	60 × 100 × 6
Tube offset from transom	32
Blocks	32 × 20 × 44 (x2)

Material: ASA, 6 walls

- Plate bolts to the transom with 4 x M5 316 bolts, bedded in sealant over a glass-reinforced pad.
- Tube bores are teardrop-shaped so they print without support.
- M4 set screws (D3.3 tap holes) lock the tube in both blocks.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

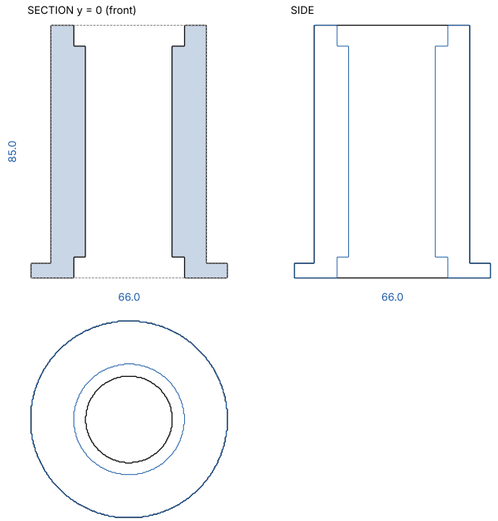
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D11

REV A

P07 Mast bearing housing

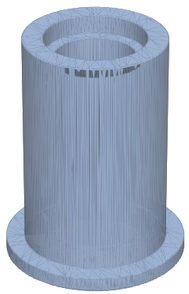
File: stl/P07_mast_bearing_housing.stl Views at 1:2 Print orientation: z up from the bed



Feature	Size
OD	52.3
Flange	D66 × 5
Length	85
Bearing seats	2 x D37.05 × 7

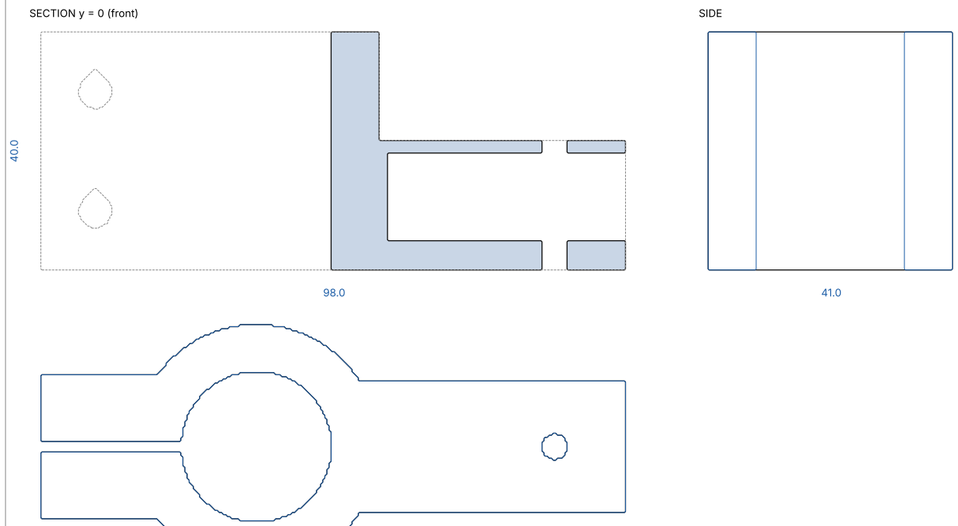
Material: PETG or ASA, 100% infill

- Bonds into the top of the 2 in. sch 40 PVC mast tube (ID 52.5); flange rests on the tube end.
- Two 6805-2RS stainless bearings (25×37×7), 78 mm apart, carry the 25 mm carbon wing spar.
- Printed flange-down as modelled.



P08 Spar clamp with tail-boom socket

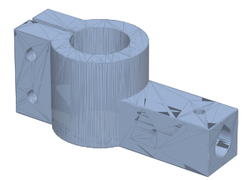
File: stl/P08_spar_clamp_boom_socket.stl Views at 1:1 Print orientation: z up from the bed



Feature	Size
Collar	OD41, ID25.2 × 40
Boom socket	D12.2 × 40
Ears	20 × 24

Material: ASA or PETG, 100% infill

- Clamps the 25 mm spar just below the wing root (2 x M5 bolts through the ears).
- Boom: 12 mm carbon tube, 40 mm deep in the socket, pinned with an M4 bolt (D4.3 vertical hole).
- Boom axis is 11 mm above the clamp's bottom face.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

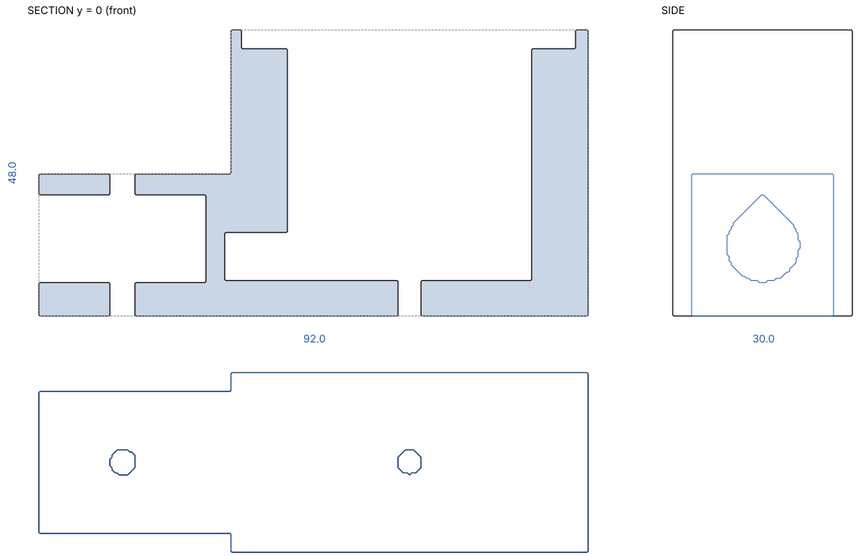
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D12

REV A

P09 Tail servo housing (boom end)

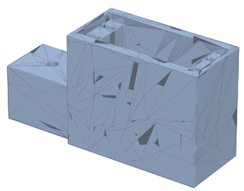
File: stl/P09_tail_servo_housing.stl Views at 1:1 Print orientation: z up from the bed



Feature	Size
Overall	92 × 30 × 48
Servo pocket	41.2 × 21.2 × 42
Boom socket	D12.2 × 28

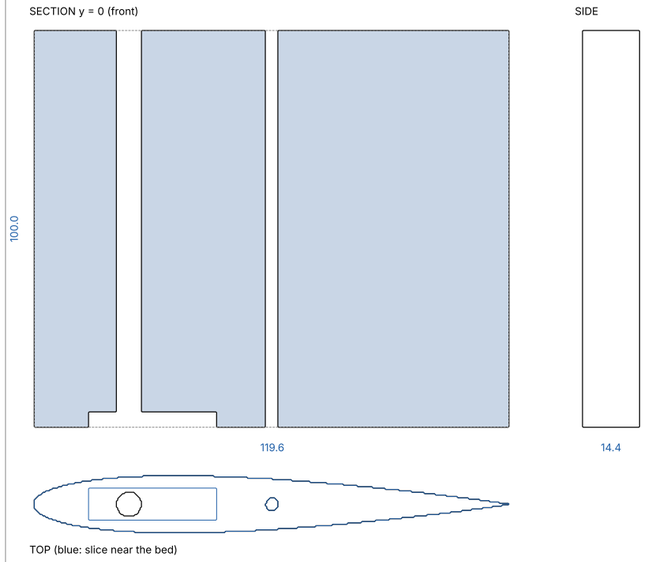
Material: ASA or PETG

- Slides on the aft end of the 12 mm boom; pin with M4 (D4.3 hole).
- Holds a Savox SW-1210SG output-up; the tail fin P10 bonds to the servo horn.
- Tab screw slots allow 47.5-51.5 mm tab spacing: check your servo before printing.
- Pocket has a D4 drain hole; cable exits through the slot toward the boom.



P10 Tail fin, lower and upper halves [tail fin lower]

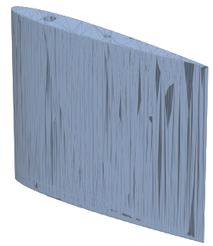
File: stl/P10a_tail_fin_lower.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Chord	120
Span	2 × 100
Thickness	14.4
Pivot	D6.2 at 24 from LE

Material: PETG or ASA, 0.2 mm layers

- NACA 0012, 120 mm chord, 200 mm span in two 100 mm halves printed standing up.
- D6.2 bore at 24 mm from the leading edge takes a 6 mm carbon pivot rod through both halves.
- D3.2 bore at 60 mm takes a 3 mm carbon rod to align and stiffen the joint.
- Lower half: 32 × 8 × 4 recess on the bottom face to bond an aluminium 25T servo horn.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

Date 2026-11-01 Drawn with Claude Check all dims before cutting

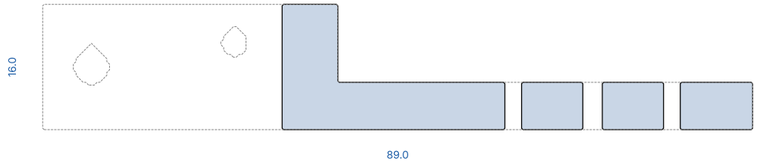
SHEET D13

REV A

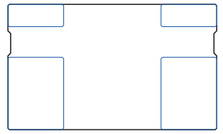
P11 Rudder tiller arm

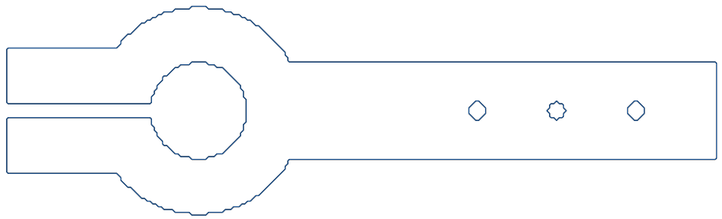
File: stl/P11_tiller_arm.stl Views at 1:0.75 Print orientation: z up from the bed

SECTION y = 0 (front)



SIDE



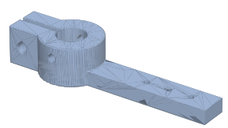


TOP (blue: slice near the bed)

Feature	Size
Hub	D26 × 16
Arm	65 × 12 × 6
Stock bore	D12.1

Material: PETG, 100% infill

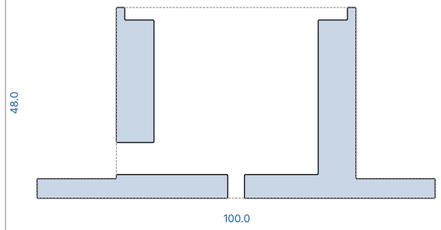
- Clamps the 12 mm 316 rudder stock (M4 bolt through the ears).
- Cross-drill the stock 3 mm through the D3.2 hole and fit a 316 pin.
- Pushrod holes (D2.2) at 35, 45 and 55 mm from the stock give three throw settings.



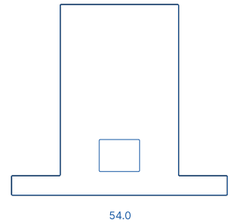
P12 Rudder servo mount

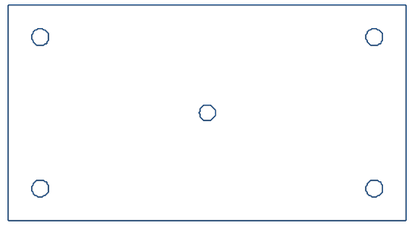
File: stl/P12_rudder_servo_mount.stl Views at 1:1.5 Print orientation: z up from the bed

SECTION y = 0 (front)



SIDE



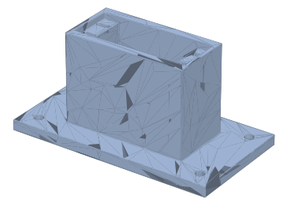


TOP (blue: slice near the bed)

Feature	Size
Base	100 × 54 × 5
Cradle	60 × 30 × 48

Material: PETG or ASA

- Bonds to a glass-reinforced pad on the hull floor beside the rudder tube (4 x M4 or bond + screw).
- Holds a Savox SW-1210SG output-up, level with the tiller arm.
- Tab screw slots allow 47.5-51.5 mm tab spacing: check your servo before printing.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

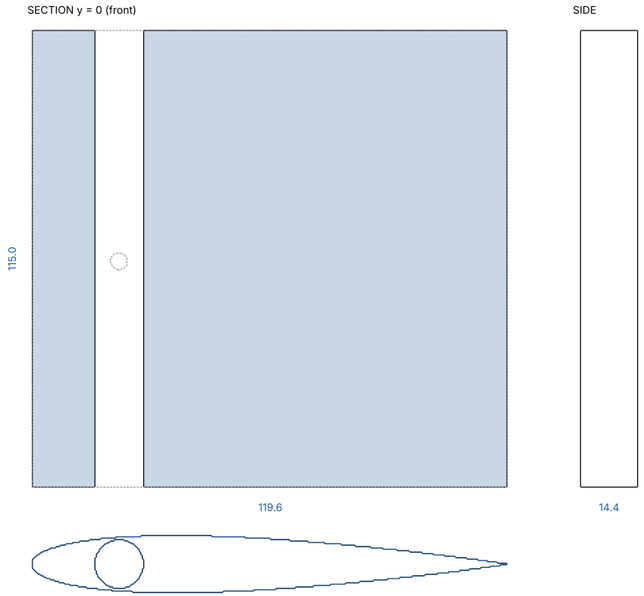
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D14

REV A

P13 Rudder blade core, upper and lower halves [rudder core upper]

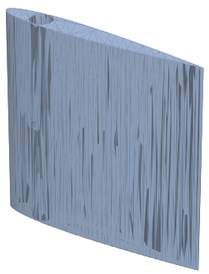
File: stl/P13a_rudder_core_upper.stl Views at 1:1.5 Print orientation: z up from the bed



Feature	Size
Chord	120
Span	2 × 115
Stock bore	D12.2 at 22 from LE

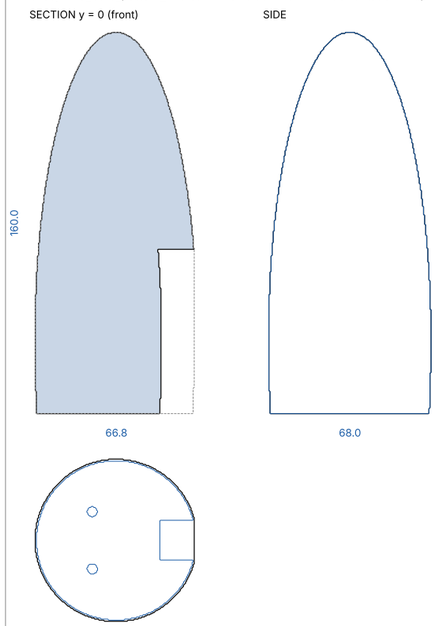
Material: PETG, 15% gyroid

- NACA 0012, 120 mm chord; each half 115 mm, printed standing up.
- Slide both halves on the 12 mm 316 stock, pin each with a 4 mm 316 pin (D4.2 holes), fill the joint with epoxy.
- Sand, then laminate 2 layers of 6 oz glass over the whole blade.
- Stock sits 22 mm behind the leading edge (about 15% balance area).



P14 Keel bulb casting pattern, forward and aft halves [bulb pattern fwd]

File: stl/P14a_bulb_pattern_fwd.stl Views at 1:2.5 Print orientation: z up from the bed



Feature	Size
Length	320
Max diameter	68
Fin slot	150 × 18 × 15

Material: PLA or PETG, 3 walls

- Pattern for a plaster or sand mould: do NOT pour lead into the print.
- Halves split at 160 mm; join with two 4 mm dowels (D4.2 holes) and fill/sand smooth.
- Slot on top matches the fin tip (NACA 0012, 150 mm chord), 15 mm deep.
- Finished lead bulb: about 8.2 kg; allow 1-2% for shrinkage when sizing the mould.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

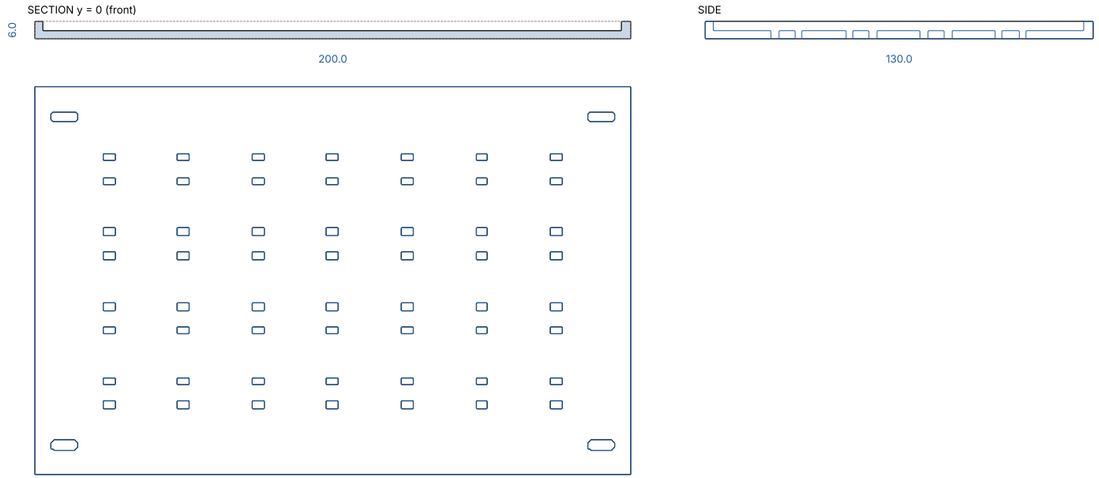
Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D15

REV A

P15 Electronics tray

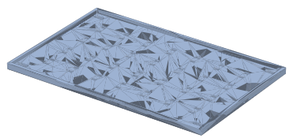
File: stl/P15_electronics_tray.stl Views at 1:2 Print orientation: z up from the bed



Feature	Size
Plate	200 × 130 × 3
Rim	3 high
Zip slots	4 × 2.5 on a 25 mm grid

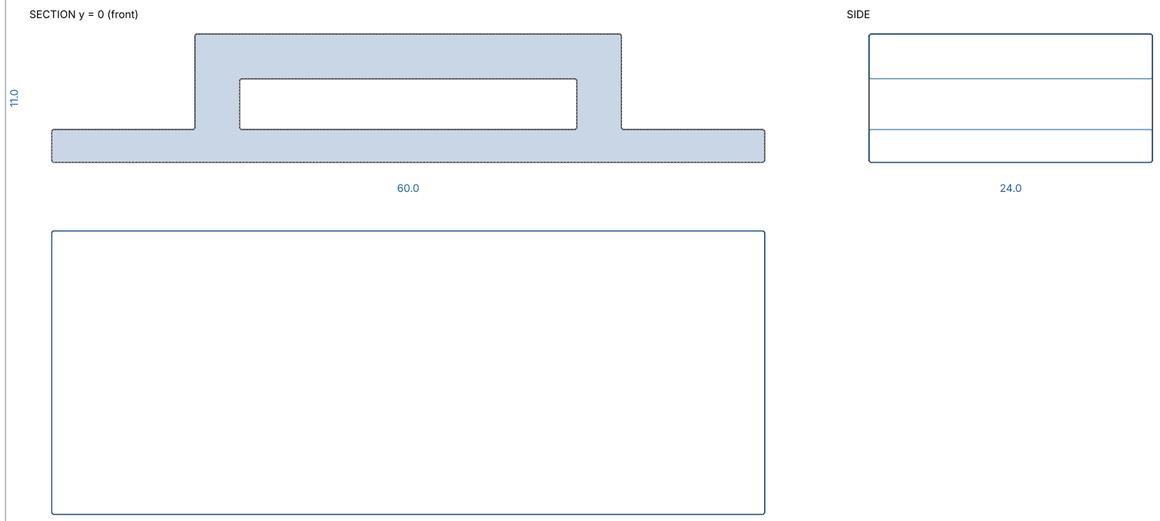
Material: PETG

- Sits inside the IP68 enclosure; trim the outline to your enclosure's inside size.
- Mount boards with 3M Dual Lock and zip ties through the 4 × 2.5 mm slot pairs (no hole patterns are assumed).
- Corner slots take M3 screws into the enclosure's mounting bosses.



P16 Battery strap anchor (qty 4)

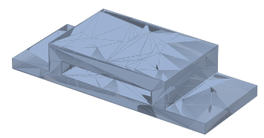
File: stl/P16_battery_strap_anchor_x4.stl Views at 1:0.5 Print orientation: z up from the bed



Feature	Size
Base	60 × 24 × 3
Webbing slot	28 × 4

Material: PETG

- Bond to the hull floor with thickened epoxy around the battery; 25 mm webbing passes under the bridge.
- Use two straps with cam buckles; strap tension holds the battery against capsize loads.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D16

REV A

P17 Penetrator boss pad (qty 4)

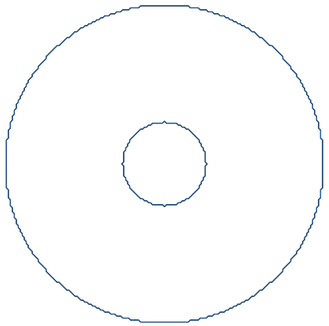
File: stl/P17_penetrator_pad_x4.stl Views at 1:0.75 Print orientation: z up from the bed

SECTION y = 0 (front)



SIDE



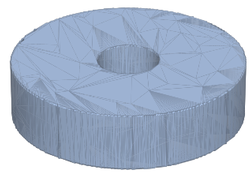


TOP (blue: slice near the bed)

Feature	Size
OD	40
Height	10
Hole	D10.5

Material: ASA or PETG, 100% infill

- Gives each Blue Robotics WetLink M10 penetrator a flat, thick seat on the curved deck or hull.
- Bond with thickened epoxy, drill through the hull using the D10.5 hole as a guide.



SOLAR-SAIL OCEAN DRONE 1.5 m

3D-PRINTED PARTS

Units: mm Views to scale as noted per part

Date 2026-11-01 Drawn with Claude Check all dims before cutting

SHEET D17

REV A